

Course III: Corporate Finance

Summer Term 2026

Unit 5 Mini-Cases

Please upload your solutions and indicate the Mini-Cases you have solved before 21:59 on April 19, 2026. There will be no deadline extensions!

To upload your solutions and make indications, please go to the assignment section on Canvas and choose “Unit 5 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). Only complete submissions containing answers to the T/F questions and a solutions upload will be considered. Your last complete attempt counts.

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solution file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Mini Case 5.1. A company can invest in a new project with an investment cost of 42 and a certain cash flow of 54. The company has no internal resources and must raise external financing.

The current asset value of the company (before the project) is either 290 or 140, each with probability 0.5. All participants are risk-neutral, the risk-free rate is zero, and managers maximize the wealth of existing shareholders.

- a. What share of post-investment company value does the manager have to offer to new external investors so that they finance the investment cost of 42?
- b. If the manager wants to raise the investment cost with debt instead of equity, what repayment do debtholders require?
- c. Does the manager raise new equity, new debt, or refrain from investing in the new project?

Now assume that the company has outstanding debt with a face value of 170. Assume that newly raised debt for the project would be *junior* to the existing debt.

- d. What share of post-investment *equity* value does the manager have to offer to new external investors so that they finance the investment cost of 42?
- e. If the manager wants to raise the investment cost with debt instead of equity, what repayment do new junior debtholders require?
- f. Does the manager raise new equity, new debt, or refrain from investing in the new project?
- g. Which classical corporate finance problem is described in this case? Briefly discuss potential solutions.

Mini Case 5.2. Consider the firm of Mini Case 5.1: parts d–g (same baseline numbers).

The government considers two possible policies:

A direct subsidy, which provides a transfer of 12 to the firm in all states of the world;

a credit guarantee, which guarantees the repayment of newly issued *junior debt* if the firm cannot repay.

- a. Under the direct subsidy, compute post-investment payoffs to equity in both states and determine whether the manager chooses to invest.
- b. Under the credit guarantee, explain how new junior creditors are protected and determine whether the firm can raise the 42 at fair terms.
- c. Compare the two policies in terms of (i) effectiveness in overcoming debt overhang and (ii) fiscal cost to the government. Which policy is more efficient?

Mini Case 5.3. Consider the same company as in Mini Case 5.1: parts a–c (no outstanding debt). Assume the manager knows that the true current asset value (without the investment) is 290. However, she cannot credibly disclose this information. Investors believe that the asset value is either 290 (Good) or 140 (Bad) with equal probability.

- a. What share of post-investment company value does the manager have to offer to new external investors so that they finance the investment cost of 42?
- b. If the manager wants to raise the investment cost with debt instead of equity, what repayment do debtholders require?
- c. Does the manager raise new equity or new debt to finance the project? What objective is she maximizing?

Now assume instead that the manager knows the true current asset value is 140, while investors still believe both values are equally likely.

- d. What share of post-investment company value does the manager have to offer to new external investors so that they finance the investment cost of 42?

- e. If the manager wants to raise the investment cost with debt instead of equity, what repayment do debtholders require?
- f. Does the manager raise new equity or new debt to finance the project?
- g. What signal does new equity issuance convey to the market in this setting?

Mini Case 5.4. A large Japanese corporation operates a single business line and generates uncertain operating profits. The risk-free rate is zero. Corporate income taxes are levied at the firm level under the following tax regime:

- Effective tax rate: 30%
- Net operating losses (NOLs) can be carried forward for up to 10 years.
- NOL carryforwards may offset at most 50% of taxable income in any positive year.

The firm's operating profits over the next two years are given by:

$$\pi_1 = \begin{cases} 500 & \text{with probability } \frac{1}{2}, \\ -400 & \text{with probability } \frac{1}{2}, \end{cases} \quad \pi_2 = 300 \text{ with certainty.}$$

The firm can implement a risk management strategy that eliminates all uncertainty in year 1. Under this strategy, year 1 operating profit is equal to its expected value with certainty. The risk management strategy has zero direct cost.

- a. Compute the firm's expected tax payments and after-tax operating profits in year 1 and year 2 *without* risk management.
- b. Compute the firm's tax payments and after-tax operating profits in each year *with* the risk management strategy.
- c. Compare expected after-tax firm value with and without risk management. Does risk management increase firm value? Explain precisely which features of the Japanese tax system drive your result.
- d. Suppose instead that individual shareholders can hedge the firm's year 1 operating risk in financial markets at zero cost. Would such shareholder-level hedging affect the firm's tax payments or firm value? Explain.
- e. From year 3 to 11 the firm just broke even. However, in year 12 it was able to generate a sizable profit of 500. How high is the firm's taxable income in year 12
 - i. *with* risk management in year 1,
 - ii. *without* risk management in year 1?

Mini Case 5.5. A firm has outstanding zero-coupon debt with face value 2,400, due in one year. No interim debt service is required. The risk-free interest rate is zero. The firm's existing assets will generate a certain cash flow of 2,050 in one year. The firm is considering whether to undertake a new investment project.

The project requires an upfront investment of 300 today and will generate a certain cash flow of 520 in one year.

Assume limited liability for equity holders.

- a. Compute the *net present value (NPV)* of the project.
- b. Will equity holders be willing to provide the required 300 to finance the project? Explain your reasoning by comparing equity payoffs with and without the project.
- c. Suppose debt holders agree to reduce the total face value of debt to 2,200.
 - i. Will equity holders now be willing to finance the project?
 - ii. Is it optimal for debt holders, as a group, to accept this reduction in face value?
- d. There are 1,200 identical bondholders, each holding a bond with face value 2.
 - i. If one bondholder believes that all other bondholders will reduce the face value of their bonds proportionally, does she want to reduce the face value of her own bond?
 - ii. If one bondholder believes that no other bondholders will reduce the face value of their bonds, does she want to reduce the face value of her own bond?
- e. Briefly discuss what this example illustrates about *debt overhang* and the role of coordination among creditors.

Mini Case 5.6. A manager has control over a publicly listed company with 2 million shares outstanding and a market capitalization of 40 million. The firm is entirely equity-financed. The manager can divert funds to projects she likes but which destroy shareholder value. If she does *not* divert funds, the market value of equity remains at 40 million. If she *does* divert funds, the market value of equity falls by 3 million.

The share price is noisy and therefore the firm has decided against using performance pay. Individual shareholders can monitor the manager. If the manager is not monitored, she diverts funds with probability 50%. If the manager is monitored, diversion is detected with certainty and does not occur. If diversion is detected, the company can sue and recover the 3 million. However, monitoring is costly: hiring a corporate governance expert for a year costs 300,000.

- a. Suppose ownership is perfectly concentrated, i.e. one shareholder owns all shares. Is it optimal for this shareholder to monitor the manager? (Assume the manager observes whether she is monitored before deciding whether to divert funds.)

- b. Suppose ownership is perfectly dispersed, i.e. each shareholder holds only one share. Is it optimal for an individual shareholder to monitor the manager?
- c. Suppose ownership is intermediate: A blockholder owns $x > 1$ shares while the remaining shares are held by small shareholders who each hold one share. When is it optimal for the blockholder to monitor the manager?
- d. Provide intuition for your results in parts (a)–(c).